

Scalability & Future Plan

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Team ID: SWTID-2026-8694
Project Name: AI BlogNest API
Maximum Marks: 1 Mark

Scalability & Future Plan

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	Local/server resource limitations	May affect response time as user traffic increases.	High
2	Increasing blog and user data	Larger datasets may require more storage and efficient queries.	Medium
3	Single-server dependency	A single server may become a bottleneck during high traffic.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Suitable for the current project usage.	Use load balancing, horizontal scaling and cloud deployment for higher traffic.
2	Data Storage	Stores project data in the current database setup.	Use scalable database infrastructure, indexing and backup strategies.
3	Performance	API performance depends on server resources and request load.	Add caching, optimize API/database queries and monitor response times.
4	Security	API access is protected through the current application setup.	Strengthen authentication, authorization, validation and security monitoring.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Cloud deployment and API monitoring	Short term	Improved availability and monitoring.
Phase 2	Caching and database optimization	Medium term	Better response time and data handling.
Phase 3	Horizontal scaling and load balancing	Medium term	Supports increased concurrent users.
Phase 4	Advanced AI features and analytics	Long term	Extends project capabilities and user value.